

Simulation-based verification of **bfpr**

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This report compares **bfpr** predictions with simulations across the listed priors, evidence thresholds, and sampling schedules.

1 z -Test

Simulated Design Priors

Table 1: Simulated design priors for the z -Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	USD
Z1	Point(0)	10 to 500 by 10	10000	1
Z2	Point(0.2)	10 to 500 by 10	10000	1
Z3	Point(0.5)	10 to 500 by 10	10000	1
Z4	Normal(0.5, 0.1)	10 to 500 by 10	10000	1
Z5	Normal(0.2, 0.3)	10 to 500 by 10	10000	1
Z6	Normal(0.2, 0.5)	10 to 500 by 10	10000	1
Z7	Normal(0, 0.5)	10 to 500 by 10	10000	1
Z8	Point(-0.3)	10 to 500 by 10	10000	1
Z9	Point(0)	10 to 500 by 10	10000	50
Z10	Point(10)	10 to 500 by 10	10000	50
Z11	Normal(10, 20)	10 to 500 by 10	10000	50
Z12	Point(0)	10 to 10000 by 10	10000	1
Z13	Point(0.2)	10 to 10000 by 10	10000	1
Z14	Normal(0.2, 0.1)	10 to 10000 by 10	10000	1
Z15	Normal(0.2, 0.5)	10 to 10000 by 10	10000	1

Analysis Priors

Table 2: Analysis priors used for the z -Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
bf01	Normal ₊ (-0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal ₊ (0, 0.1)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal ₊ (0, 0.707)	Point(0)	Z1-Z5, Z7-Z8, Z12-Z13
bf01	Normal ₊ (0, 2)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal ₊ (0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8

bf01	Normal ₋ (-0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal ₋ (0, 0.1)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal ₋ (0, 0.707)	Point(0)	Z1-Z5, Z7-Z8, Z12-Z13
bf01	Normal ₋ (0, 2)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal ₋ (0.2, 0.3)	Point(0)	Z1-Z5, Z7-Z8
bf01	Normal(0, 0.707)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0, 20)	Point(0)	Z9-Z11
bf01	Normal(0.2, 0.3)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0.2, 0.707)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Normal(10, 20)	Point(0)	Z9-Z11
bf01	Point(0.2)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Point(10)	Point(0)	Z9-Z11
dirbf01	Normal ₊ (0, 0.707)	Normal ₋ (0, 0.707)	Z1-Z8, Z12-Z15
dirbf01	Normal ₊ (0, 20)	Normal ₋ (0, 20)	Z9-Z11
dirbf01	Normal ₊ (0.2, 0.707; > 0.2)	Normal ₋ (0.2, 0.707; ≤ 0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(0.354)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(7.071)	Point(0)	Z9-Z11

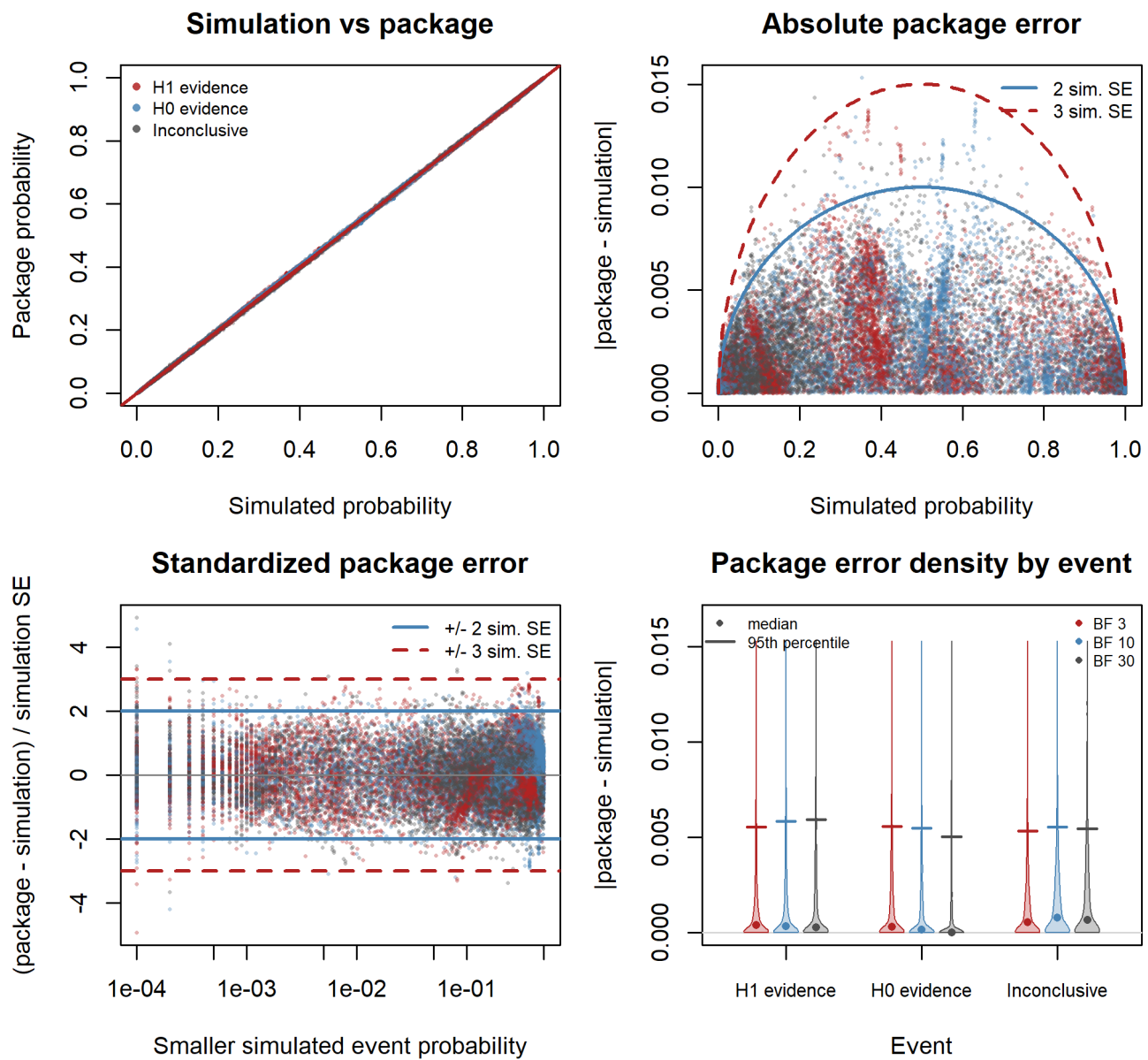
Normal(μ, σ) uses mean μ and SD σ before truncation. Subscripts $+$ and $-$ denote truncation above and below zero; a nonzero cutoff is shown explicitly. Normal design priors remain untruncated.

1.1 Fixed-Design

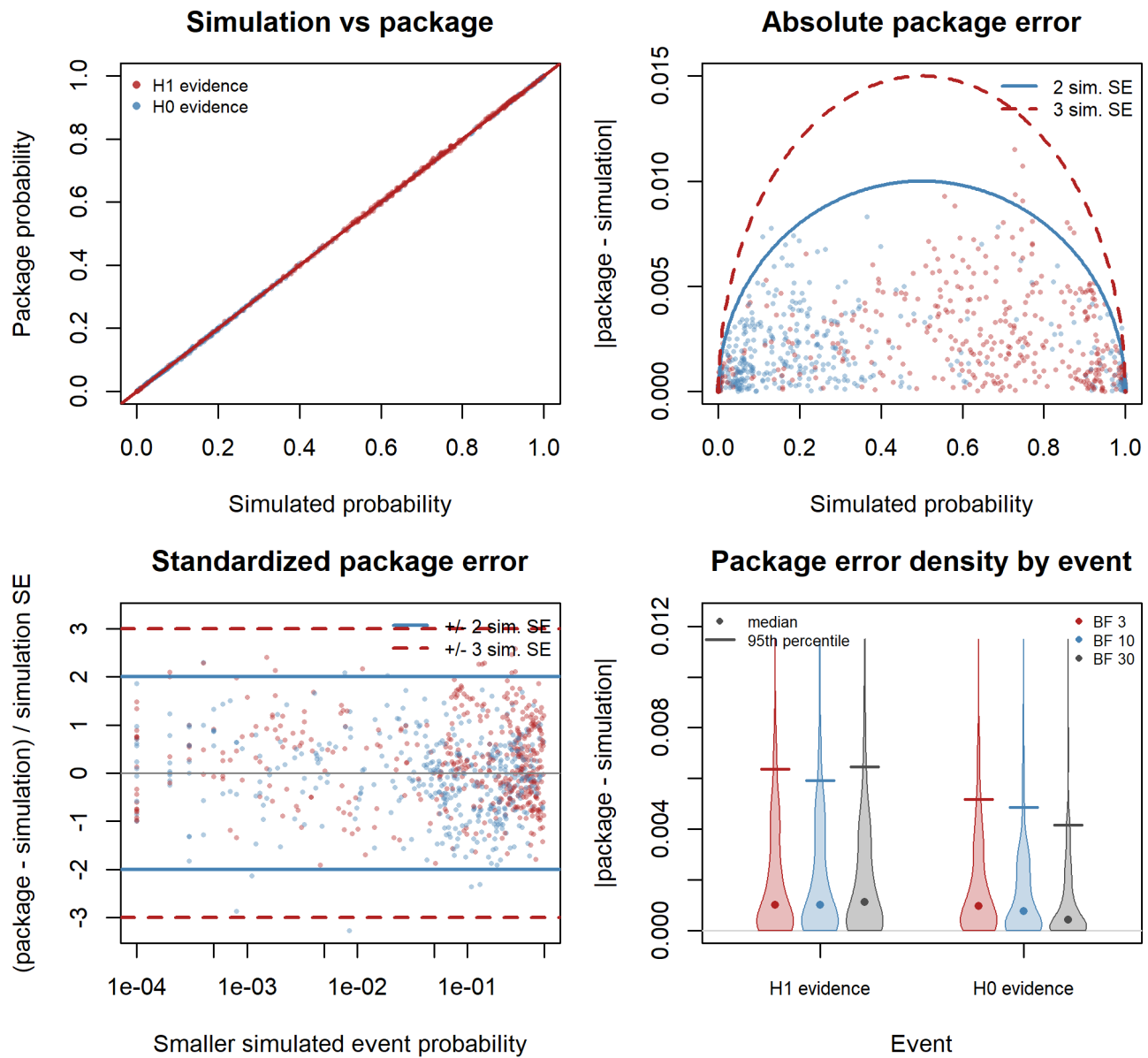
Table 3: Fixed-design verification settings for z-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Sample sizes
bf01	51	Z1-Z11	BF 3, 10, 30	30, 130, 260, 380, 500
bf01	22	Z12-Z15	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
nmbf01	24	Z1-Z11	BF 3, 10, 30	30, 130, 260, 380, 500
nmbf01	11	Z12-Z15	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
dirbf01	16	Z1-Z11	BF 3, 10, 30	30, 130, 260, 380, 500
dirbf01	7	Z12-Z15	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
bf01 (Normal ₊ , Normal ₋)	70	Z1-Z5, Z7-Z8	BF 3, 10, 30	10 to 500 by 10

bf01 fixed-design checks



nmbf01 fixed-design checks



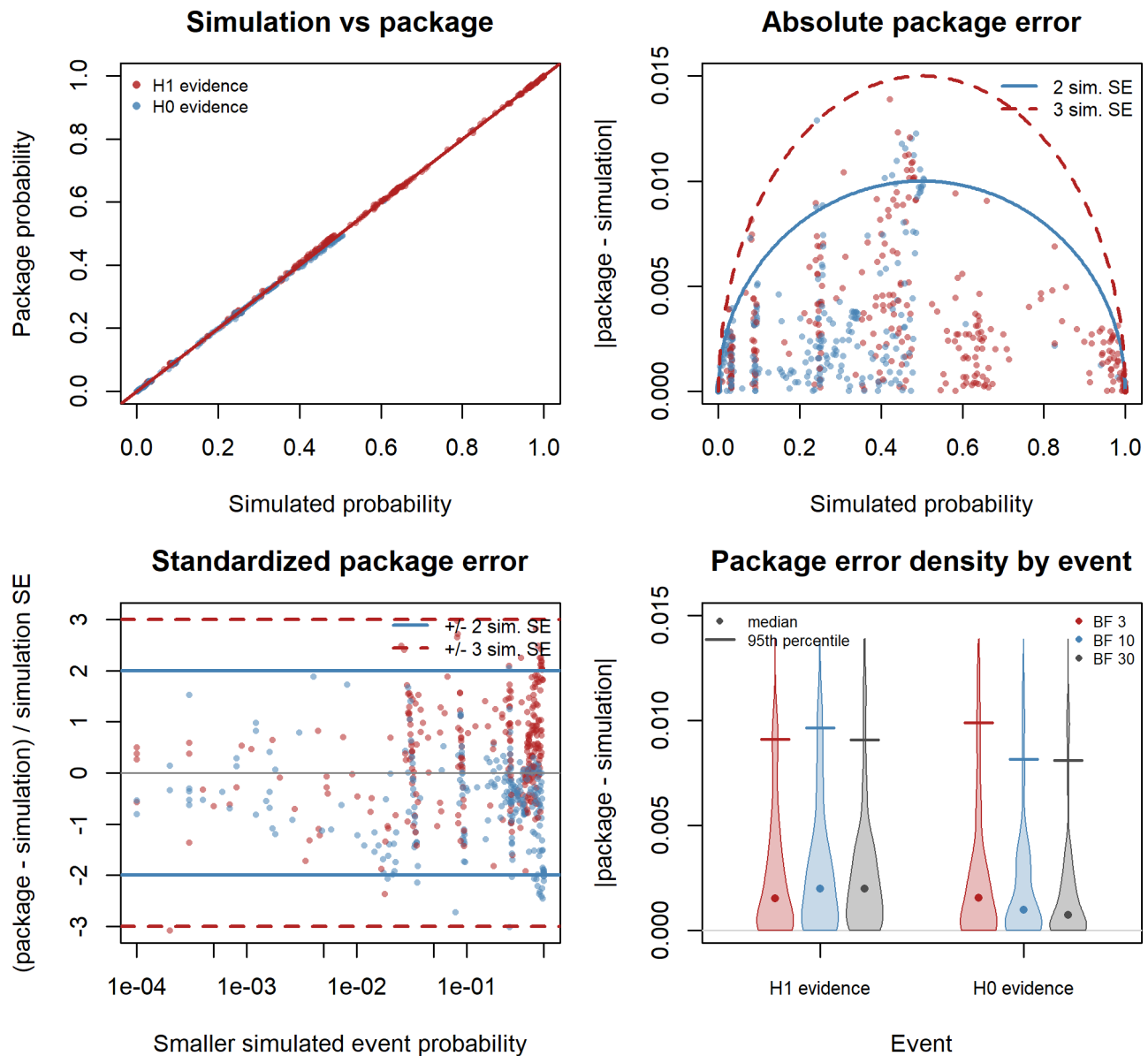
dirbf01 fixed-design checks**1.2 Sequential-Design**

Table 4: Sequential-design verification settings for z-Test analyses. Designs refers to the design-prior table.

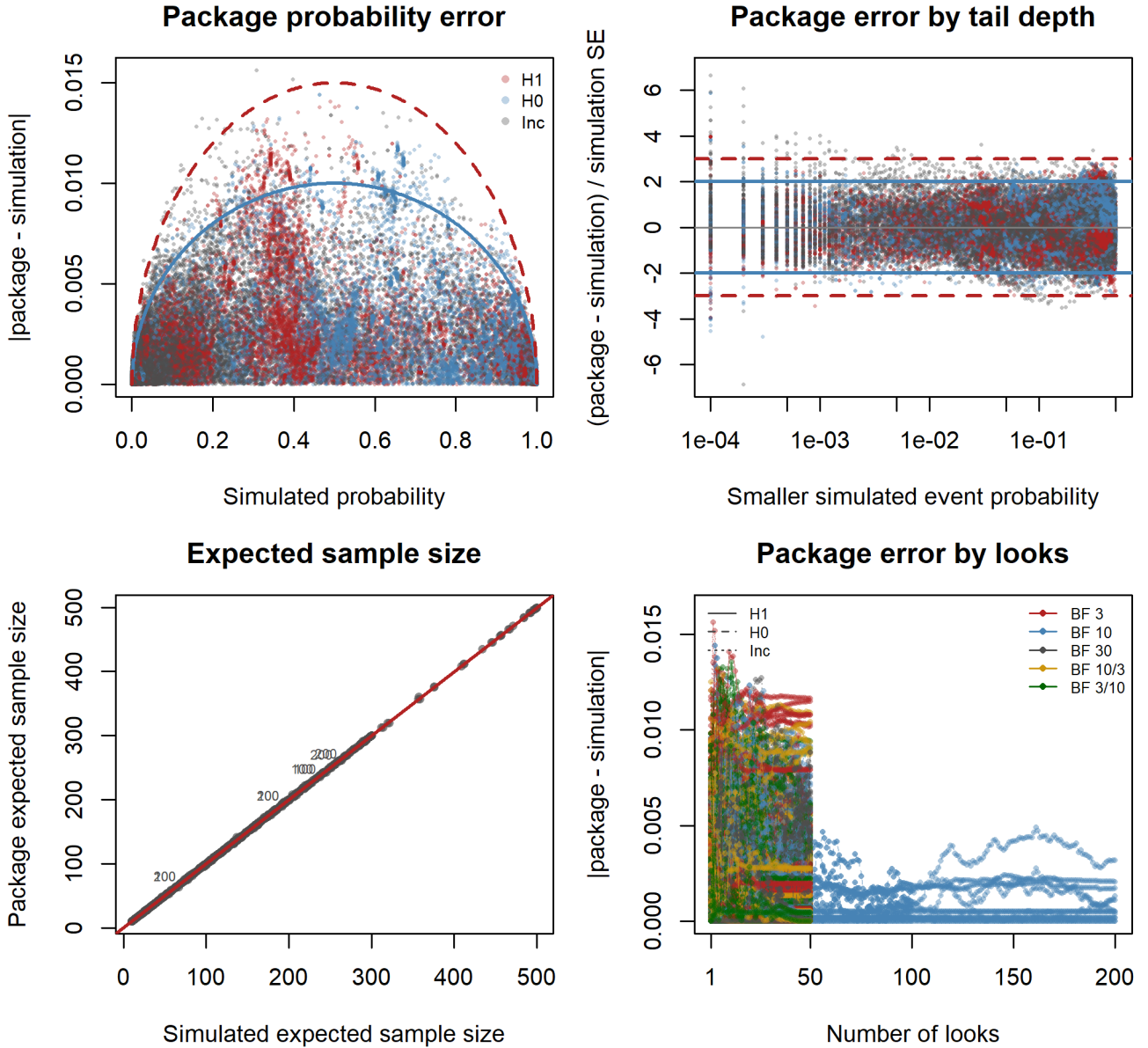
Function	Prior/design pairs	Designs	Evidence thresholds	Look sequence
bf01	16	Z4-Z5, Z7, Z11-Z12, Z14	BF 3, 10, 30	start 20, by 10 (2, 5, 10 looks)
nmbf01	16	Z4-Z5, Z7-Z8, Z11-Z12, Z14	BF 3, 10, 30	start 20, by 10 (2, 5, 10 looks)
dirbf01	11	Z4-Z5, Z7-Z9, Z11-Z12, Z14	BF 3, 10, 30	start 20, by 10 (2, 5, 10 looks)
bf01 (Normal ₊ , Normal ₋)	74	Z1-Z5, Z7-Z8, Z12-Z13	BF 3, 10, 30; 10/3, 3/10	1, 3, 50 looks; selected 100, 200 looks

The Normal₊ and Normal₋ cases use a single look at 300, three equally spaced looks (100, 200, 300), three uneven looks (20, 100, 300), and 50 looks from 10 to 500. For the centered prior with SD 0.707 and point design priors at 0 and 0.2, schedules also contain 100 or 200 looks in steps of 10. BF 10/3 and BF 3/10 denote asymmetric stopping thresholds $(k_1, k_0) = (1/10, 3)$ and $(1/3, 10)$.

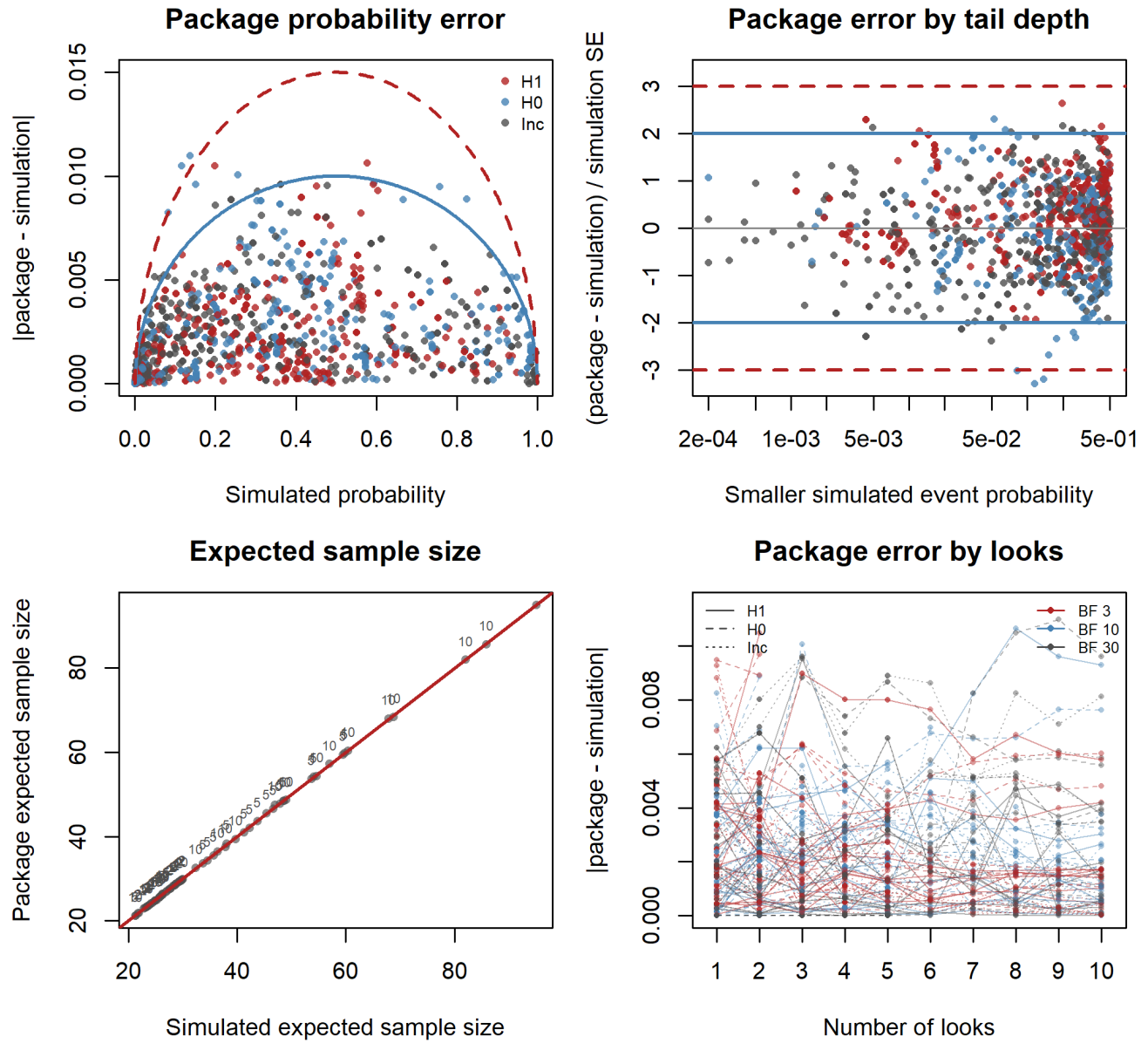
Very rare stopping probabilities can be missed by the integration grid. Events whose fixed-look union bound implies fewer than one expected occurrence in 10,000 trajectories are retained as accuracy diagnostics, including their original interval failures. Their relative accuracy is not certified; larger discrepancies and invalid probabilities remain required failures.

For example, the Normal₋(0,0.1) analysis prior and Normal(0,0.5) design prior with $(k_1, k_0) = (1/30, 30)$ give an H0 stopping prediction of 2.22×10^{-19} by $n = 110$, whereas the fixed-look union bound is 1.18×10^{-5} . The simulation contains 1 such event in 10,000 trajectories.

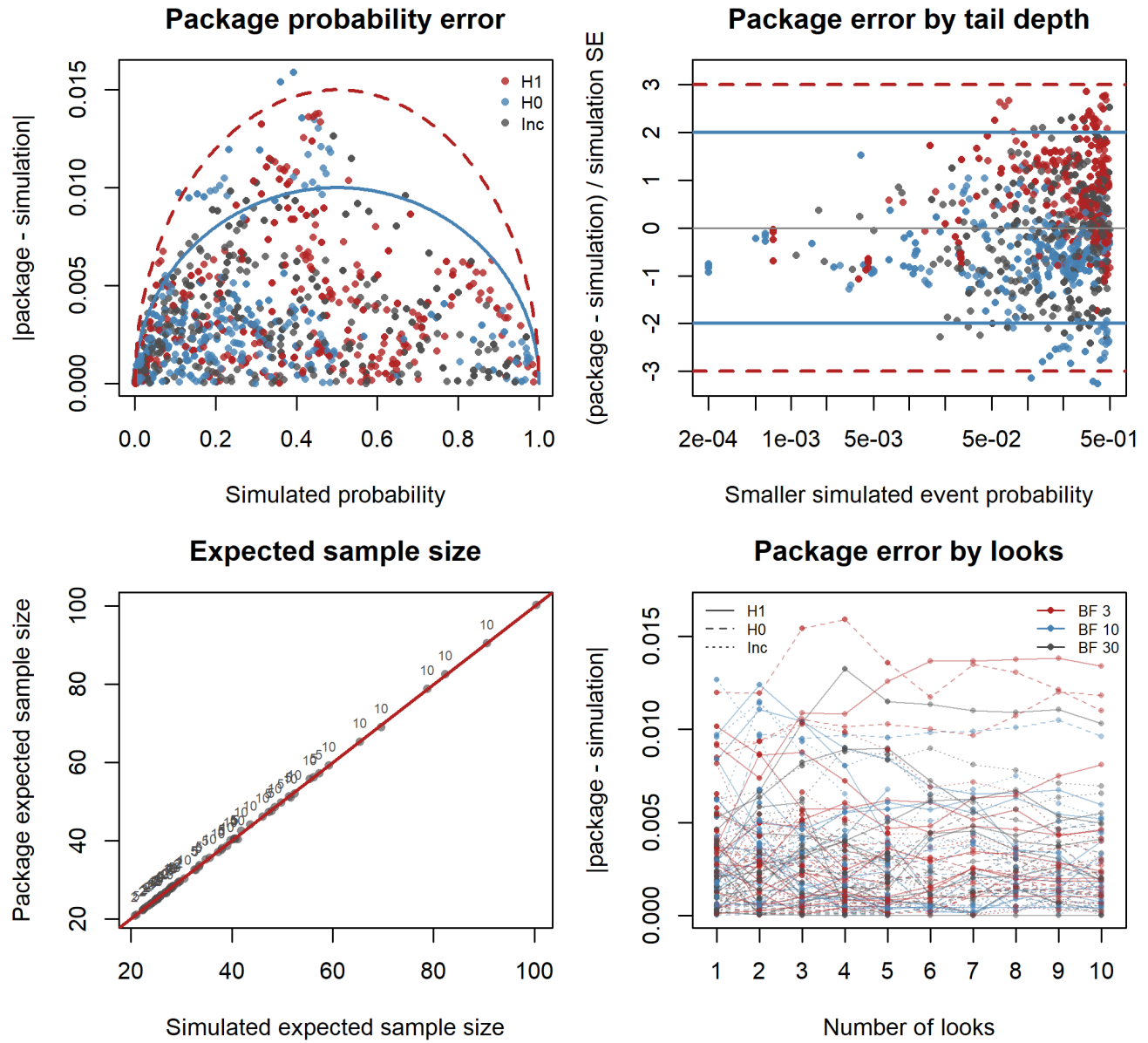
bf01 sequential-design checks



nmbf01 sequential-design checks



dirbf01 sequential-design checks



2 *t*-Test

Simulated Design Priors

Table 5: Simulated design priors for the *t*-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	Sampling	n2/n1
T1	Point(0)	10 to 500 by 10	10000	two-sample	1
T2	Point(0.5)	10 to 500 by 10	10000	two-sample	1
T3	Normal(0.5, 0.1)	10 to 500 by 10	10000	two-sample	1
T4	Normal(0.5, 0.5)	10 to 500 by 10	10000	two-sample	1
T5	Point(0)	10 to 500 by 10	10000	one-sample	
T6	Point(0.5)	10 to 500 by 10	10000	one-sample	
T7	Point(0)	10 to 500 by 10	10000	paired	
T8	Point(0.5)	10 to 500 by 10	10000	paired	
T9	Point(-0.4)	10 to 500 by 10	10000	two-sample	1
T10	Point(0)	10 to 500 by 10	10000	two-sample	1.1
T11	Point(0.35)	10 to 500 by 10	10000	two-sample	1.1
T12	Normal(0, 0.5)	10 to 500 by 10	10000	two-sample	1
T13	Point(0)	10 to 10000 by 10	10000	two-sample	1
T14	Point(0.2)	10 to 10000 by 10	10000	two-sample	1
T15	Normal(0.2, 0.1)	10 to 10000 by 10	10000	two-sample	1
T16	Normal(0.2, 0.5)	10 to 10000 by 10	10000	two-sample	1

Analysis Priors

Table 6: Analysis priors used for the *t*-Test verification run. Designs lists the matching short IDs from the design-prior table.

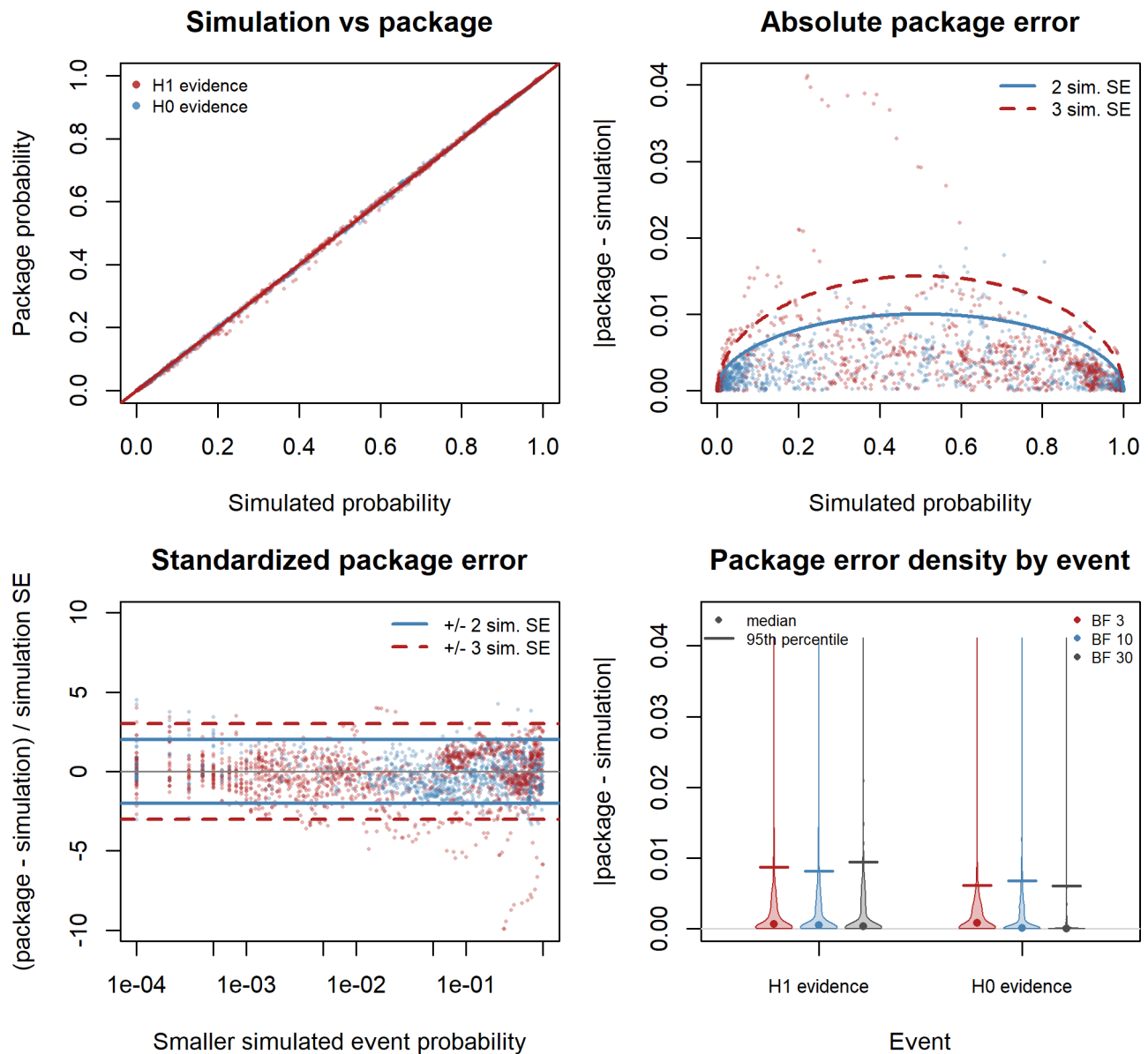
Function	H1	H0	Designs
tbff01	Cauchy ₊ (0, 0.707)	Point(0)	T1-T16
tbff01	Cauchy ₋ (0, 0.707)	Point(0)	T1-T16
tbff01	Cauchy(0, 0.35)	Point(0)	T1-T16
tbff01	Cauchy(0, 0.707)	Point(0)	T1-T16
tbff01	Cauchy(0, 1)	Point(0)	T1-T16
tbff01	StudentT ₊ (3, 0.5, 0.1)	Point(0)	T2-T4, T6, T8, T11, T14-T16
tbff01	StudentT ₊ (3, 0.5, 0.35; > 0.2)	Point(0.2)	T11, T14-T16
tbff01	StudentT ₊ (30, 0.5, 0.35; > 0.2)	Point(0.2)	T1, T5, T7, T10, T13
tbff01	StudentT(3, 0, 0.707)	Point(0)	T1-T16
tbff01	StudentT(30, 0, 0.707)	Point(0)	T1-T16

2.1 Fixed-Design

Table 7: Fixed-design verification settings for t-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Sample sizes
tbf01	95	T1-T12	BF 3, 10, 30	30, 130, 260, 380, 500
tbf01	35	T13-T16	BF 3, 10, 30	510, 2510, 5000, 7500, 10000

tbf01 fixed-design checks



2.2 Sequential-Design

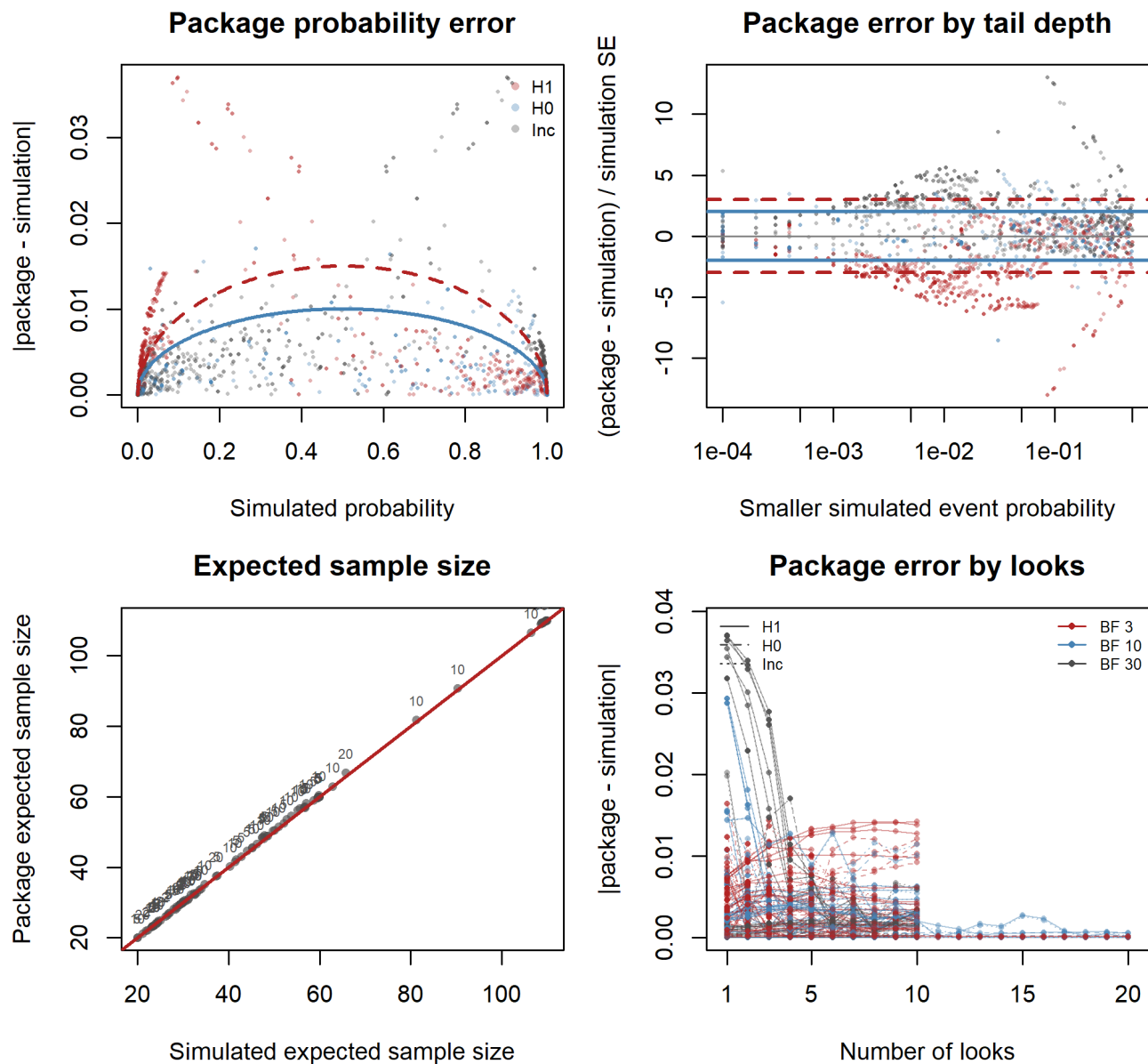
Table 8: Sequential-design verification settings for t-Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Look sequence
tbf01	27	T2, T5-T6, T9, T13	BF 3, 10, 30	start 20, by 10 (2, 5, 10, 20 looks)

The sequential *t*-test probabilities use a normal approximation to the sampling distribution of the test statistics. Comparisons with exact-*t* simulations therefore diagnose this approximation; they do not certify exact-*t* accuracy. Numerical validity and fixture integrity remain required checks.

In this run, 199 of 2688 probability comparisons fall outside the severe binomial prediction intervals; the largest absolute difference is 3.703 percentage points.

tbf01 sequential-design checks



3 Binomial Test

Simulated Design Priors

Table 9: Simulated design priors for the Binomial Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions
B1	Point(0.5)	10 to 500 by 10	10000
B2	Point(0.6)	10 to 500 by 10	10000
B3	Point(0.2)	10 to 500 by 10	10000
B4	Point(0.75)	10 to 500 by 10	10000
B5	Beta(60, 40)	10 to 500 by 10	10000
B6	Beta ₊ (1, 1; > 0.5)	10 to 500 by 10	10000
B7	Beta ₋ (1, 1; ≤ 0.5)	10 to 500 by 10	10000
B8	Point(0.5)	10 to 10000 by 10	10000
B9	Point(0.55)	10 to 10000 by 10	10000

Analysis Priors

Table 10: Analysis priors used for the Binomial Test verification run. Designs lists the matching short IDs from the design-prior table.

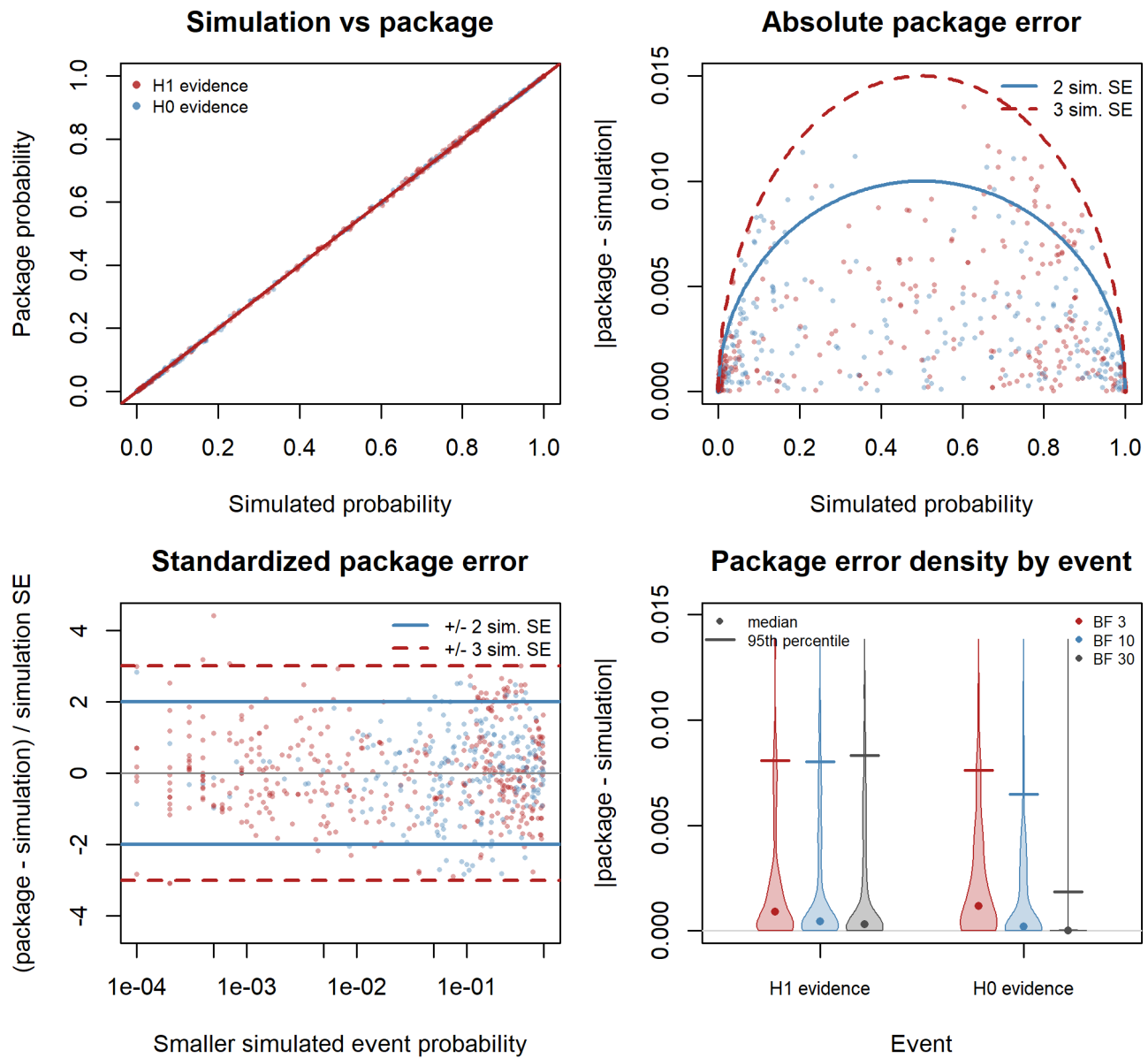
Function	H1	H0	Designs
binbf01 (direction)	Beta ₊ (1, 1; > 0.2)	Beta ₋ (1, 1; ≤ 0.2)	B3
binbf01 (direction)	Beta ₊ (1, 1; > 0.5)	Beta ₋ (1, 1; ≤ 0.5)	B1-B9
binbf01 (direction)	Beta ₊ (1, 1; > 0.55)	Beta ₋ (1, 1; ≤ 0.55)	B9
binbf01 (direction)	Beta ₊ (1, 1; > 0.6)	Beta ₋ (1, 1; ≤ 0.6)	B2, B5
binbf01 (direction)	Beta ₊ (1, 1; > 0.75)	Beta ₋ (1, 1; ≤ 0.75)	B4
binbf01 (point)	Beta(1, 1)	Point(0.2)	B3
binbf01 (point)	Beta(1, 1)	Point(0.5)	B1-B9
binbf01 (point)	Beta(1, 1)	Point(0.55)	B9
binbf01 (point)	Beta(1, 1)	Point(0.6)	B2, B5
binbf01 (point)	Beta(1, 1)	Point(0.75)	B4
binbf01 (point)	Beta(40, 60)	Point(0.5)	B1-B9
binbf01 (point)	Beta(5100, 4900)	Point(0.5)	B8-B9
binbf01 (point)	Beta(60, 40)	Point(0.5)	B1-B9

3.1 Fixed-Design

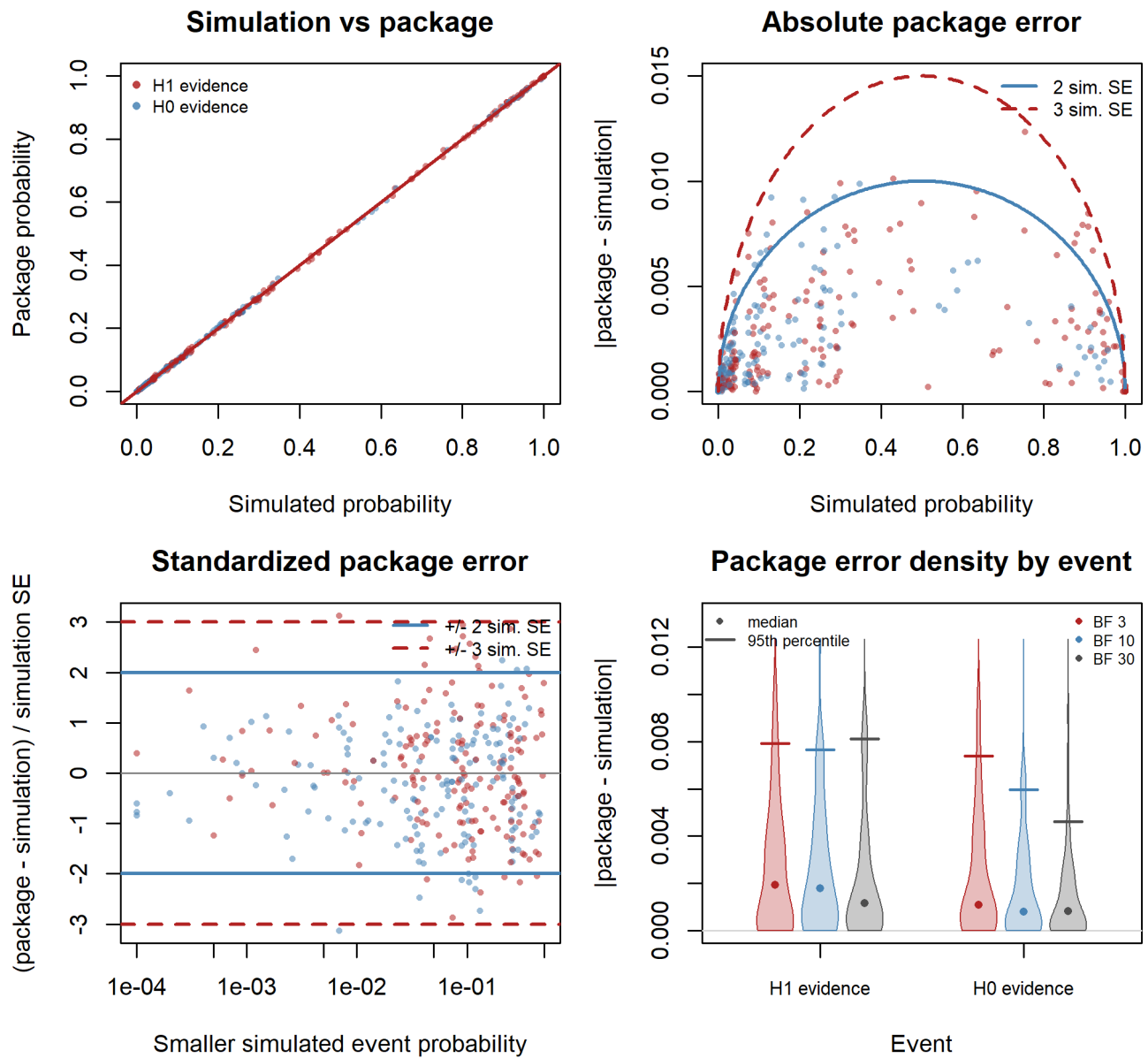
Table 11: Fixed-design verification settings for Binomial Test analyses. Designs refers to the design-prior table.

Function	Prior/design pairs	Designs	Evidence thresholds	Sample sizes
binbf01 (point)	25	B1-B7	BF 3, 10, 30	30, 130, 260, 380, 500
binbf01 (point)	9	B8-B9	BF 3, 10, 30	510, 2510, 5000, 7500, 10000
binbf01 (direction)	11	B1-B7	BF 3, 10, 30	30, 130, 260, 380, 500
binbf01 (direction)	3	B8-B9	BF 3, 10, 30	510, 2510, 5000, 7500, 10000

binbf01-point fixed-design checks



binbf01-direction fixed-design checks



4 Sample-Size Search Checks

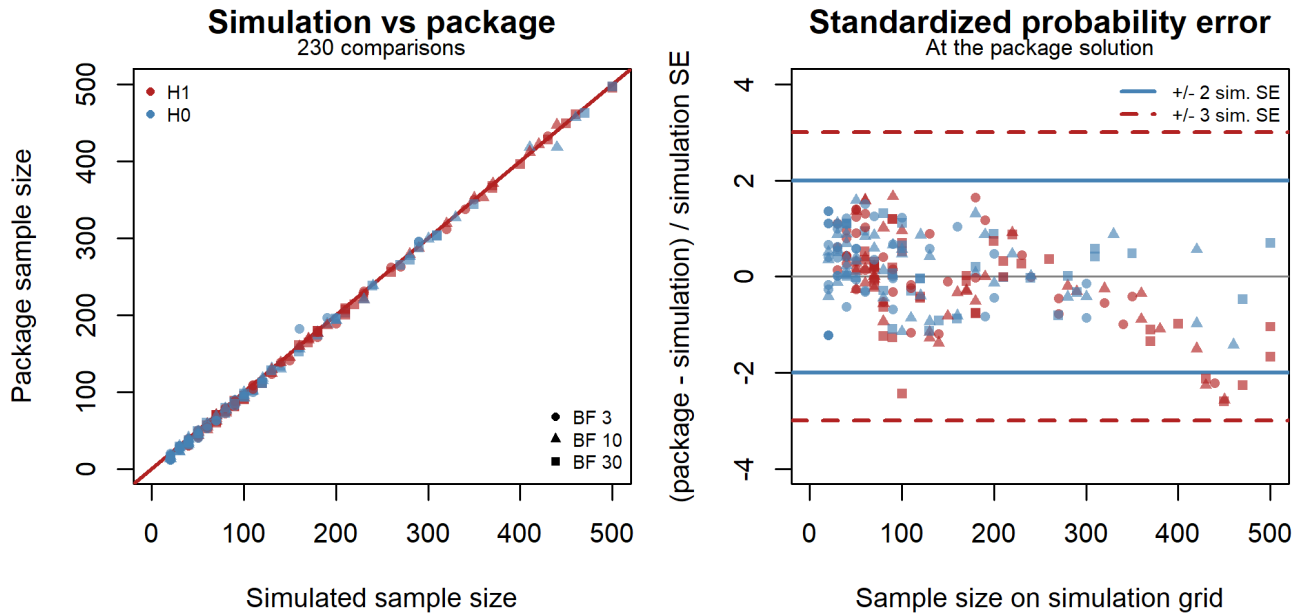
The searches cover `nbf01` with one-sided Normal priors, `nbf01seq`, and `ntbf01seq`. They compare the first sample sizes at which simulated and computed probabilities of evidence for H1 or H0 reach a target. Sequential sample sizes refer to the maximum planned sample size. Only finite results from both methods enter the figures.

Left panels compare sample sizes. Right panels compare computed and simulated probabilities at the package solution, expressed in Monte Carlo SEs, with reference lines at ± 2 and ± 3 SEs. For fixed designs, the package solution is rounded up to the simulation grid for the probability comparison.

4.1 Fixed-Design z-Test: `nbf01`

These comparisons cover the 70 Normal₊ and Normal₋ analysis-prior/design-prior combinations listed in the z-test section. Targets are 0.8 and 0.9 for both H1 and H0, with BF thresholds 3, 10, and 30 and a search range from 10 to 500. The package searches over integer sample sizes; simulated probabilities are available every 10 observations.

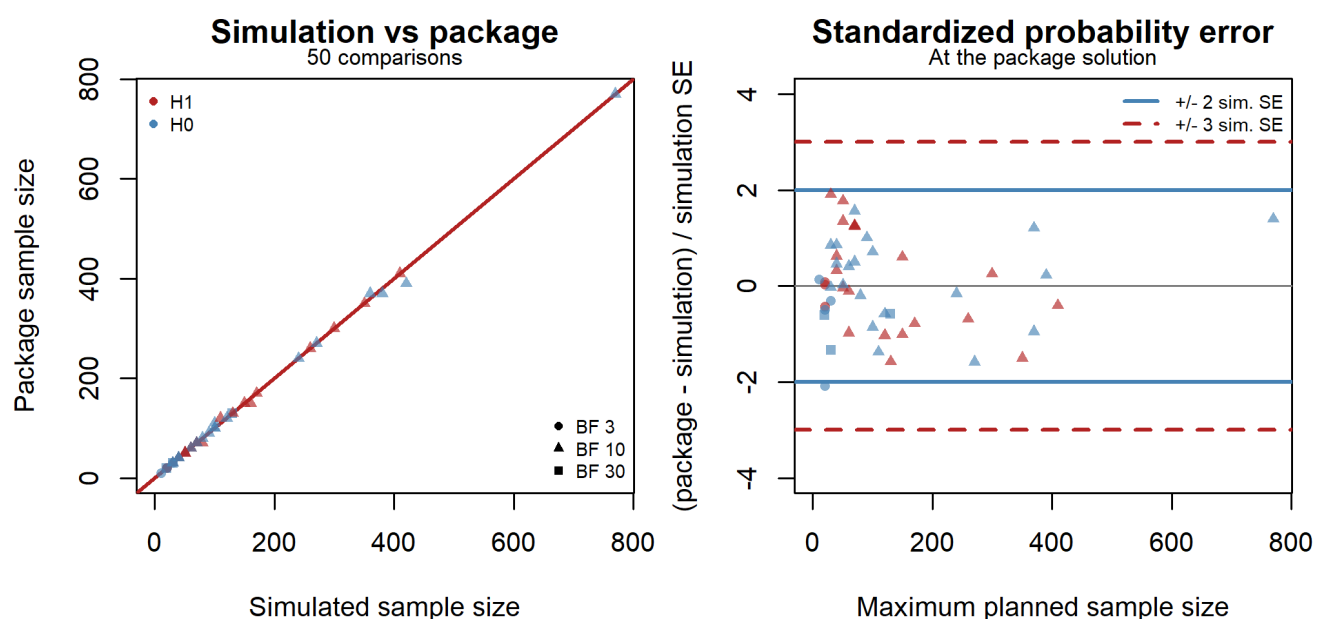
`nbf01`: one-sided normal



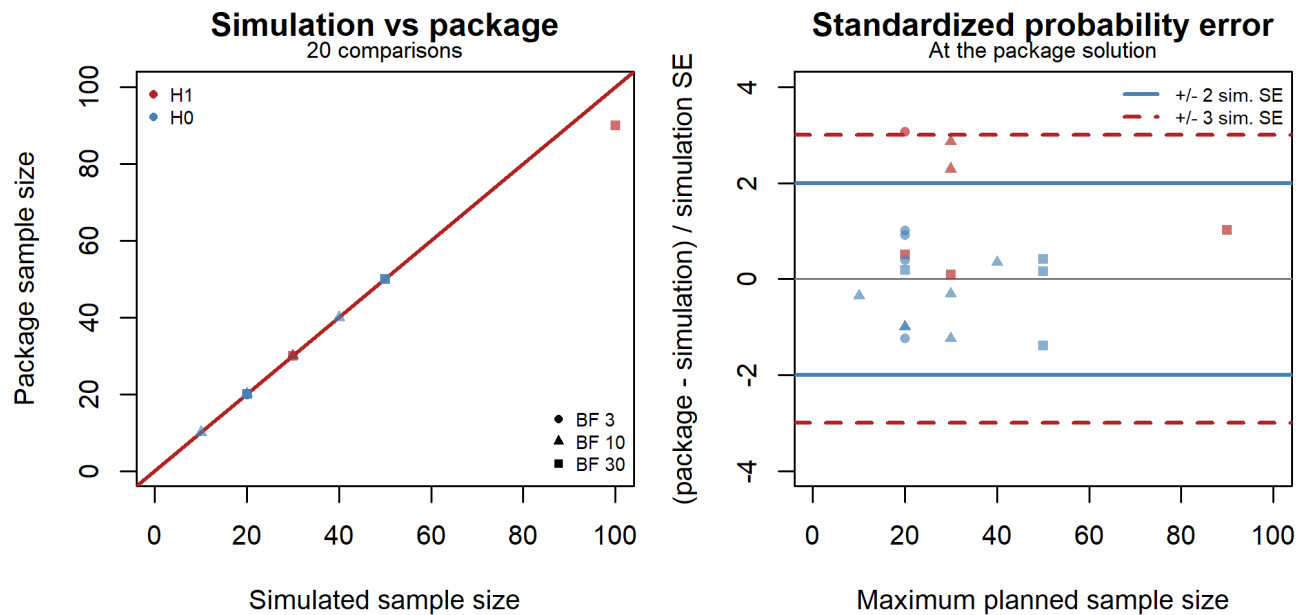
4.2 Sequential z-Test: nbf01seq

The normal and point priors, normal moment priors, and directional hypotheses use 15, 20, and 15 searches, respectively, with the analysis and design priors listed in the z-test section. Targets are 0.1, 0.3, and 0.8 for all three models, and 0.95 for directional hypotheses. BF thresholds are 3, 10, and 30 for H1 and H0. Looks occur every 10 observations starting at 10 or 20. Upper search bounds depend on the scenario and extend to 2,000 for normal and directional models and 1,000 for moment priors. The one-sided normal cases use $\text{Normal}_+(0, 0.707)$, $\text{Normal}_-(0, 0.707)$, $\text{Normal}_+(0.2, 0.3)$, and $\text{Normal}_-(0.2, 0.3)$, all seven generating distributions, and targets 0.8 and 0.9 for H1 and H0. These 112 searches use $(k_1, k_0) = (1/10, 10)$ and looks from 10 to 500 in steps of 10.

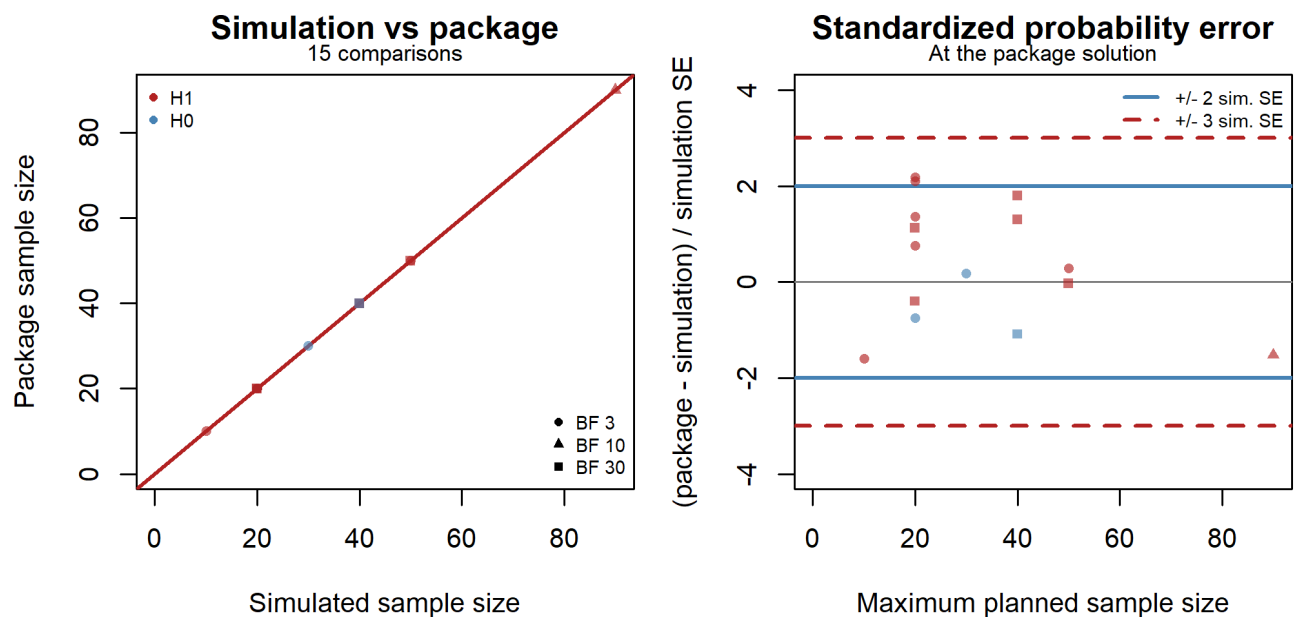
nbf01seq: normal and point priors



nbf01seq: normal moment prior



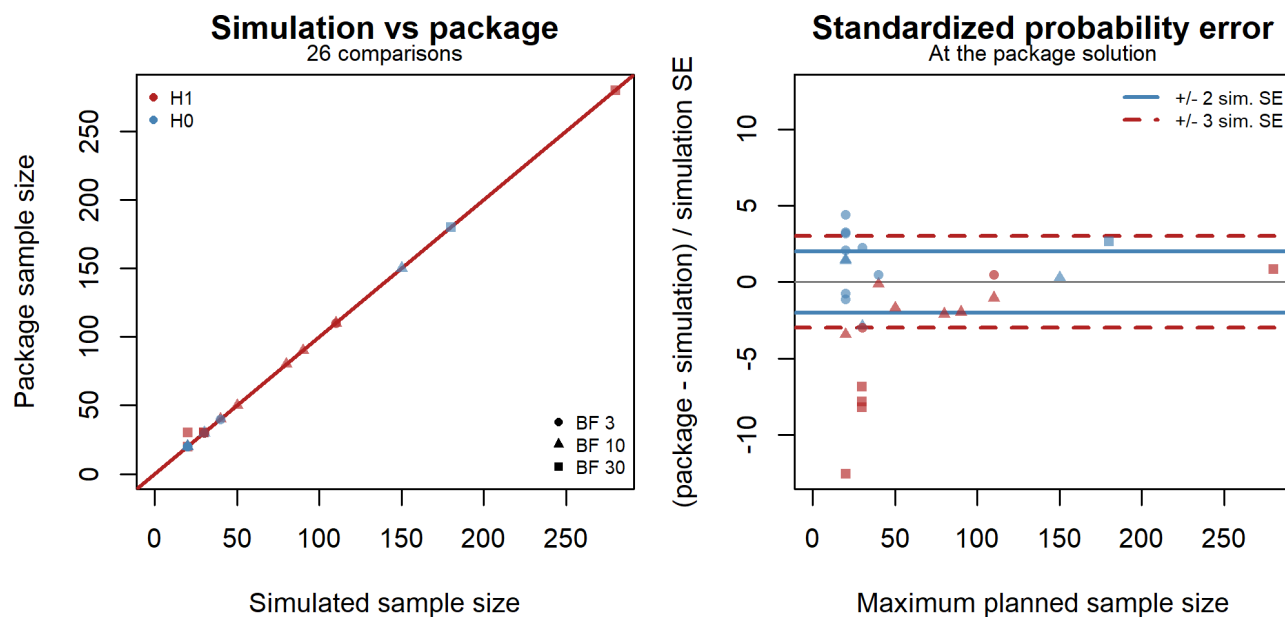
nbf01seq: directional hypotheses



4.3 Sequential t-Test: *ntbf01seq*

The 26 searches use Cauchy and StudentT analysis priors with two-sided, greater, and less alternatives, and point and Normal design priors. Sampling is one-sample, paired, or two-sample, with two-sample allocation ratios 1 and 1.1. Targets are 0.1, 0.3, 0.8, and 0.95, with BF thresholds 3, 10, and 30 for H1 and H0. Looks occur every 10 observations starting at 10 or 20, and upper search bounds range from 30 to 2,000. Two-sample sizes refer to group 1. For allocation ratio 1.1, group-2 sizes are rounded to the nearest integer in the simulation and upward in the package search. The sequential t probabilities use a normal approximation to the sampling distribution of the test statistics.

ntbf01seq: t-test priors



5 Run Time Diagnostics

Elapsed times are recorded for the package calculations shown above. Fixed-design probability calls can cover several rows at once because those functions are vectorized; most sequential and sample-size search calls are one setting at a time.

Calculations used 6 concurrent workers.

Table 12: Probability and sample-size search run times. Fixed probability rows can be vectorized, so calls and rows covered need not match.

Function	Calls	Rows covered	Median	P90	Max	Total
pbf01	73	2,190	0 s	0.02 s	0.11 s	0.42 s
pbf01seq	239	906	0.08 s	2.596 s	1.494 min	23.873 min
pbinbf01	48	1,440	0.06 s	0.08 s	0.31 s	2.97 s
pnmbf01	35	1,050	0 s	0.02 s	0.3 s	0.43 s
ptbf01	130	3,900	1.37 s	2.631 s	3.14 s	3.008 min
ptbf01seq	148	148	0.565 s	3.575 s	1.463 min	12.684 min
nbf01seq	50	50	0.01 s	0.513 s	1.63 min	1.966 min
ntbf01seq	26	26	0.175 s	3.585 s	11.94 s	31.38 s

Table 13: One-sided z-test run times. Each fixed pbf01 run times the H1 and H0 vector calls together; each other run times one package call.

Function	Timed runs	Median	P90	Max	Total
nbf01	840	0 s	0.02 s	0.07 s	3.21 s
nbf01seq	112	8.46 s	10.479 s	11.7 s	12.338 min
pbf01	210	0.025 s	0.04 s	0.08 s	5.23 s
pbf01seq	1408	0.03 s	9.78 s	1.935 min	64.066 min

Histograms show the calls in the probability and sample-size search timing table.

Package Function Call Timing

